

Model Analysis Report

Building Resilience Index Model

MKM-BRI-001

Report Date: 15 May 2026

Governance Framework: SR 11-7 / SS1/23 Model Risk Management

Purpose

This analysis report supplements the main Model Documentation (`bri_resilience.pdf`) with sensitivity studies that examine how the flood resilience score responds to perturbations of key inputs. The full mathematical framework, assumptions, and limitations are documented in the main paper; this report focuses purely on quantitative sensitivity.

Sensitivity Study Design

All sensitivities use the same baseline property archetype: a 1976–1999 Thames property in Zone 2 with Fair condition, Fluvial regime, and Meets minimum compliance on every measure. One input is varied at a time while others are held fixed. Each row reports the mean final flood resilience score across 200 samples with matched random seeds.

Sensitivity Tables

Table 1: Mean flood resilience score by construction period

Period	Mean score (1-100)
Pre-1919	58.6
1919-1944	54.8
1945-1975	45.2
1976-1999	63.1
2000-2008	75.0
2009-Present	83.1

Table 2: Mean flood resilience score by EA flood zone

Zone	Mean score (1-100)
Zone 1	58.4
Zone 2	63.1
Zone 3a	69.9
Zone 3b	74.4

Table 3: Mean flood resilience score by property condition

Condition	Mean score (1-100)
Excellent	74.4
Good	66.8
Fair	63.1
Poor	51.8
Very poor	38.3

Table 4: Mean flood resilience score by flood regime (FloodRiskType)

Regime	Mean score (1-100)
Pluvial	62.2
Fluvial	63.1
Coastal	63.1

Interpretation

The qualitative ordering matches the design intent:

- **Period** — the 1945–1975 post-war cohort scores worst, the 2009–Present cohort best. Pre-1919 stock scores middle (survivor bias plus solid construction, but no modern measures).
- **Zone** — higher EA flood zones receive a generator-side uplift on flood-protection measures, lifting their mean score. This captures the intuition that flood-zone properties are more likely to invest in resilience. Note: the model itself is not yet context-normalised by hazard intensity (see limitation BRI-L2 in the main paper).
- **Condition** — monotonic across all five condition bands. Excellent stock scores roughly 2× Very-poor stock.
- **Regime** — modest variation; the regime multipliers re-weight the same compliance values rather than changing them. A coastal-regime property with the same checklist scores within a few points of its fluvial counterpart for matched compliance.

For a full discussion of limitations, including the four proxy mappings flagged in the model documentation, see Section 8 of `bri_resilience.pdf`.